

Application for Coordinator

Horizon Centre for Innovation

Instructions

- You are very encouraged to talk to the current and previous team for any queries related to Horizon. A link to all the contacts has been provided below.
- After completing the application make it into PDF format and submit it [here](#) along with a copy of your Grade Card.
- Name the file as : "<Your Name>_<Roll Number>_CFI_Horizon_Coordinator_2025-2026"
- If you wish to provide any of your social media handles, you are free to do so.
- Although there is no solid vertical division in the club, we expect coordinators (with the help of one of the core members) to perform multiple jobs during their tenure like website, poster design, PR, and question setting for competitions. You needn't know any of these currently. We expect you to learn anything possible during the application time
- Even if you don't complete the answer, write as much as possible. You will be judged based on your learning efforts.
- The instructions under the domain sections give the number of mandatory tasks in each domain.
- Do not exceed 9 pages for the general questions
- Lastly, have fun apping and feel free to approach us for any doubts. Cheers!

Contacts

- [Horizon Team](#)

NOTE: Fret not over the length of the app! It may seem extensive. However, numerous internal options have been crafted to make your job easier. So, kick back, relax, and relish the delights of apping!

1 General Questions

1. Why do you want to become a Coordinator for the club, and what is your motivation for applying? Mention your strengths and weaknesses. Also, mention why you think you are suitable for being a coordinator.
2. What do you think will be your roles as a coordinator in the coming year? What's the difference between a Coordinator, Project Member, Guild Member, AP Team Member and a Project Manager?
3. Mention all the different events Horizon conducts and your views on them. Brownie points if you cover the most events. Do a SWOT (Strengths-Weaknesses Opportunities-Threats) analysis of the events. What changes do you want to bring in them? Use this question to lay out all your opinions and think about the reforms you'd like to bring.
4. What Projects did the club work on last year, and what is their current status? Mention possible avenues specific to Projects that the club can explore in the coming year.
5. Explain how you will increase the interaction between Horizon and other clubs and managerial teams of CFI while working as a club coordinator.
6. How widespread is the club's online presence? Do you think publicity and outreach are important for the club? How can Horizon expand and improve its online presence? Answer in detail.
7. Mention some ideas for next year's competitions. (You can come up with your novel ideas as well). What contests other than collaboration with TechSoc can we conduct? Briefly outline an event you would like to run as a coordinator. Think about and mention the issues that we could face in organising them.
8. What are your thoughts on the role of DCs? If selected as a coordinator, would you be comfortable managing and mentoring them? What kind of sessions or initiatives would you like to conduct for DCs to help them grow within the club? Additionally, if you were a DC in Horizon, please share your experience and what you learned from it.
9. Mention other PoRs you have or plan on taking up. Would becoming a coordinator leave you with less time for academics and other activities? If you do become a coordinator and face a situation where you find it difficult to cope with academics, what would you do?

2 Physics and Astronomy

2.1 Physics

Attempt atleast **two** questions

1. Take a look at [this video](#). Can you derive the Euler-Lagrange equation starting from the action principle? According to the video, does light take the single path of least action, or does it take all possible paths? If it takes all paths, why do we observe only one? Feel free to answer conceptually, mathematically, or both!
2. The Spooky Wheel Illusion – Is Time Reversing?
You're watching a thriller movie, and during a high-speed chase, you notice something strange: the wheels of a motorcycle appear to be spinning backwards. Could this be an indication of time travel, or is something else at play? Explain it mathematically.
Hint: This illusion is related to sampling theory. Think about how movies capture motion and how the frame rate might affect the perception of fast-moving objects.
What principle from sampling theory explains why the wheels appear to rotate in reverse in the movie? How does this principle relate to the frame rate at which the movie is shot?
3. Derive the generalized Uncertainty Principle. Reference material for solving this question: [Reference](#)

2.2 Astronomy

1. Investigate how the tilt of the Earth's axis influences the way sunlight reaches different parts of the Earth. How does this tilt cause phenomena such as Zero Shadow Day, the equinoxes, and seasonal changes? You may draw figures to explain your answers better.
2. Answer with respect to telescope parameters and equipment
 - (a) When looking through a telescope, what happens to an object's brightness if we magnify it? Give reason.
 - (b) What is a Barlow lens and what advantage does it give?
 - (c) What is the use of a Finderscope?
 - (d) What advantage do Cassegrain telescopes give
 - (e) What is an H-alpha filter?
3. **Toshith**, a curious young astronomer in the early 2000s known best for loafing around, loves spending his nights exploring the cosmos through his telescope. One slightly cloudy evening, he spots Sirius, Vega, and even the bright planet Jupiter shimmering between the clouds.

Intrigued by **Vega**, he reaches for an old astronomy book from the 1900s. It lists Vega's equatorial coordinates as:

- Right Ascension: $\alpha = 18^{\text{h}} 33^{\text{m}} 33.1^{\text{s}}$
- Declination: $\delta = +38^{\circ} 41' 26''$

But Toshith quickly realizes that the book is over a century old! He knows that both proper motion of stars and the precession of Earth's axis can change these values over time.

He finds the proper motion components of Vega to be:

- $\mu_{\alpha} = +0.202/\text{year}$ (in Right Ascension)
- $\mu_{\delta} = +0.286/\text{year}$ (in Declination)

Task: Help Toshith determine the updated equatorial coordinates of Vega for the epoch 2000.0, accounting for both **proper motion** and the effect of **precession** over the 100-year interval.

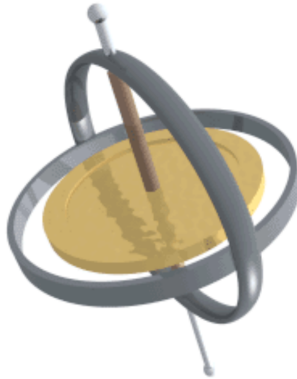


Figure 2: Precession

Things to Consider:

- Right Ascension and Declination feel fixed; why do we use them as a standard celestial coordinate system? How is this similar to or different from Altitude-Azimuth coordinates?
- If RA and Dec are defined with respect to the celestial sphere, why do they change over time? What physical phenomena drive this change?
- Could a telescope mounted with an alt-azimuth mount track stars accurately throughout the night? Why or why not?

3 Quantum Computing

Attempt atleast 2 questions

Reference material for this question: [QC Reference](#)

1. What drives the development of quantum computing, and how does its foundational approach differ from that of classical computing?
2. Why is quantum computing considered more powerful than classical computing for certain problems, and what are some real-world examples where it offers significant advantages (e.g., factoring, optimization, simulation)?
3. How are qubits defined in quantum computing, and what role does the Bloch Sphere play in visualizing their state space?
4. Please try out this coding exercise: [Google Colab Notebook](#)

4 Website

Imagine you're tasked with improving the Horizon website to make it more visually appealing, user-friendly, and easier for team members to contribute content. The current website is hosted as a static site using GitHub Pages. Please refer to current [Website's Repository](#) for context.

Task:

1. Suggest no-code or low-code platforms (e.g., Wix Studio, Framer AI, Dorik, Typedream, Notion-as-Website, etc.) that can be used to maintain and update the website.
2. Compare the pros and cons of these platforms with the current GitHub Pages-based setup.
3. Recommend UI/UX improvements that would:
 - Improve how articles and events are structured and displayed.
 - Make it easier to find past and upcoming events.
 - Allow quick and simple article uploads by coordinators/editors (without coding).
4. **(Optional):** If you try any of the suggested platforms, share a sample link of what you built.

5 Design

Attempt any **two** of the three questions

1. This is your chance to showcase your creativity! Design a poster for any of Horizon's events (you're free to choose) using an online tool or design software of your choice. We've used Canva for previous event posters, but feel free to explore other platforms as well. Along with your poster, include a brief explanation of the design approach and inspiration behind it.
(Check SMail to find some of the event posters).
2. Horizon releases a Monthly Newsletter ([Example](#)). Your task is to design a newsletter that includes the following sections:
 - Latest Research in Physics and Astronomy
 - Recent Horizon Events
 - Challenge of the Month or Riddles
 - Article of the Month
 - Your Own Creative Addition (e.g., trivia corner, member spotlight, meme, etc.)

You can refer to the existing [newsletter template](#) for format. Please create a copy of the Canva project before editing.

3. Propose a concept for a monthly Instagram reel. Refer to previous Horizon Instagram posts for inspiration. The reel can showcase recent events, invitations to Observation Sessions, or other creative ideas aligned with Horizon's vibe. Create such a short video, upload it to Drive, and paste the link in your app. Feel free to ask us for any resources you'd like, including pictures.

6 Outreach

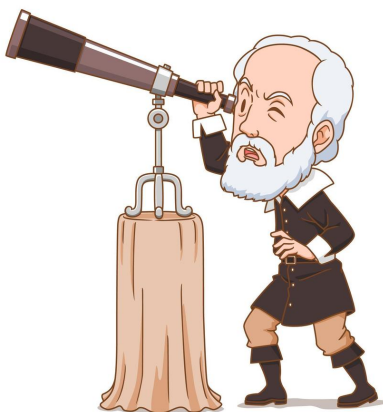
Attempt any **one** of the two questions

1. *Quark to Quasars* is one of the most remarkable sessions conducted by Horizon. Profs from various backgrounds come up with interesting talks for the GSB. Identify a professor, either from Chennai or elsewhere, whose research area you find intriguing and suitable to conduct a Q2Q session for Horizon. Draft an email introducing the club creatively and engagingly to impress the professor and request them to lead a session, preferably offline or online. Additionally, inquire about their availability and preferred dates for the session. Did you attend any Q2Q sessions? What was your experience?
2. Identify a few potential institutions or companies that Horizon could collaborate with. What mutual benefits could arise from such a partnership? Then, draft a formal email proposing a collaboration. In your email, introduce Horizon, highlight what it has to offer, and outline the proposed collaborative event. Be sure to also request sponsorship support, such as prize pools or other incentives.

7 Event Organization

Both questions are compulsory

1. Horizon conducts numerous sessions throughout the year. If given the opportunity to lead a session for Horizon, how would you make it stand out from the rest? Select your preferred topic in Physics or Astronomy and create an abstract for the session. It need not be a lecture session; you are free to choose the format.
2. Come up with specific ideas for scientific demonstrations or experiments that you would like to work on. Focus on themes related to events like Zero Shadow Day or Observation sessions where we will be able to showcase the demonstrations to the audience.



Name this curious stargazer who discovered Jupiter's largest moons